Finance and Risk Analytics Project Report

Data Science and Business Analytics - Part A

Great Learning

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1. Dataset summary and Outlier treatment

1.1 Fetch the data

Sol: Information about the analysis

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2058 entries, 0 to 2057
Data columns (total 58 columns):
 #   Column                                             Non-Null Count  Dtype
---  ------                                             --------------  -----
 0   Co_Code                                            2058 non-null   int64
 1   Co_Name                                            2058 non-null   object
 2   _Operating_Expense_Rate                            2058 non-null   float64
 3   _Research_and_development_expense_rate             2058 non-null   float64
 4   _Cash_flow_rate                                    2058 non-null   float64
 5   _Interest_bearing_debt_interest_rate               2058 non-null   float64
 6   _Tax_rate_A                                        2058 non-null   float64
 7   _Cash_Flow_Per_Share                               1891 non-null   float64
 8   _Per_Share_Net_profit_before_tax_Yuan_             2058 non-null   float64
 9   _Realized_Sales_Gross_Profit_Growth_Rate           2058 non-null   float64
 10  _Operating_Profit_Growth_Rate                      2058 non-null   float64
 11  _Continuous_Net_Profit_Growth_Rate                 2058 non-null   float64
 12  _Total_Asset_Growth_Rate                           2058 non-null   float64
 13  _Net_Value_Growth_Rate                             2058 non-null   float64
 14  _Total_Asset_Return_Growth_Rate_Ratio              2058 non-null   float64
 15  _Cash_Reinvestment_perc                            2058 non-null   float64
 16  _Current_Ratio                                     2058 non-null   float64
 17  _Quick_Ratio                                       2058 non-null   float64
 18  _Interest_Expense_Ratio                            2058 non-null   float64
 19  _Total_debt_to_Total_net_worth                     2037 non-null   float64
 20  _Long_term_fund_suitability_ratio_A                2058 non-null   float64
 21  _Net_profit_before_tax_to_Paid_in_capital          2058 non-null   float64
 22  _Total_Asset_Turnover                              2058 non-null   float64
 23  _Accounts_Receivable_Turnover                      2058 non-null   float64
 24  _Average_Collection_Days                           2058 non-null   float64

 25  _Inventory_Turnover_Rate_times                     2058 non-null   float64
 26  _Fixed_Assets_Turnover_Frequency                   2058 non-null   float64
 27  _Net_Worth_Turnover_Rate_times                     2058 non-null   float64
 28  _Operating_profit_per_person                       2058 non-null   float64
 29  _Allocation_rate_per_person                        2058 non-null   float64
 30  _Quick_Assets_to_Total_Assets                      2058 non-null   float64
 31  _Cash_to_Total_Assets                              1962 non-null   float64
 32  _Quick_Assets_to_Current_Liability                 2058 non-null   float64
 33  _Cash_to_Current_Liability                         2058 non-null   float64
 34  _Operating_Funds_to_Liability                      2058 non-null   float64
 35  _Inventory_to_Working_Capital                      2058 non-null   float64
 36  _Inventory_to_Current_Liability                    2058 non-null   float64
 37  _Long_term_Liability_to_Current_Assets             2058 non-null   float64
 38  _Retained_Earnings_to_Total_Assets                 2058 non-null   float64
 39  _Total_income_to_Total_expense                     2058 non-null   float64
 40  _Total_expense_to_Assets                           2058 non-null   float64
```

```
41  _Current_Asset_Turnover_Rate                          2058 non-null   float64
42  _Quick_Asset_Turnover_Rate                            2058 non-null   float64
43  _Cash_Turnover_Rate                                   2058 non-null   float64
44  _Fixed_Assets_to_Assets                               2058 non-null   float64
45  _Cash_Flow_to_Total_Assets                            2058 non-null   float64
46  _Cash_Flow_to_Liability                               2058 non-null   float64
47  _CFO_to_Assets                                        2058 non-null   float64
48  _Cash_Flow_to_Equity                                  2058 non-null   float64
49  _Current_Liability_to_Current_Assets                  2044 non-null   float64
50  _Liability_Assets_Flag                                2058 non-null   int64
51  _Total_assets_to_GNP_price                            2058 non-null   float64
52  _No_credit_Interval                                   2058 non-null   float64
53  _Degree_of_Financial_Leverage_DFL                     2058 non-null   float64
54  _Interest_Coverage_Ratio_Interest_expense_to_EBIT     2058 non-null   float64
55  _Net_Income_Flag                                      2058 non-null   int64

 55  _Net_Income_Flag                                     2058 non-null   int64
 56  _Equity_to_Liability                                 2058 non-null   float64
 57  Default                                              2058 non-null   int64
dtypes: float64(53), int64(4), object(1)
```

Data summary: Shape of the dataset (2058, 58)

Summary of the data:

	count	mean	std	min	25%	50%	75%	max
Co_Code	2058.0	1.757211e+04	2.189289e+04	4.000000	3.674000e+03	6.240000e+03	2.428075e+04	7.249300e+04
_Operating_Expense_Rate	2058.0	2.052389e+09	3.252624e+09	0.000100	1.578727e-04	3.330330e-04	4.110000e+09	9.980000e+09
_Research_and_development_expense_rate	2058.0	1.208634e+09	2.144568e+09	0.000000	0.000000e+00	1.994130e-04	1.550000e+09	9.980000e+09
_Cash_flow_rate	2058.0	4.652426e-01	2.266269e-02	0.000000	4.600991e-01	4.634450e-01	4.680691e-01	1.000000e+00
_Interest_bearing_debt_interest_rate	2058.0	1.113022e+07	9.042595e+07	0.000000	2.760280e-04	4.540450e-04	6.630660e-04	9.900000e+08
_Tax_rate_A	2058.0	1.147770e-01	1.524457e-01	0.000000	0.000000e+00	3.709890e-02	2.161909e-01	9.996963e-01
_Cash_Flow_Per_Share	1891.0	3.199856e-01	1.529979e-02	0.169449	3.149890e-01	3.206479e-01	3.259178e-01	4.622268e-01
_Per_Share_Net_profit_before_tax_Yuan_	2058.0	1.769673e-01	3.015730e-02	0.000000	1.666039e-01	1.756421e-01	1.858854e-01	7.923477e-01
_Realized_Sales_Gross_Profit_Growth_Rate	2058.0	2.276117e-02	2.170104e-02	0.004282	2.205831e-02	2.210001e-02	2.215200e-02	1.000000e+00
_Operating_Profit_Growth_Rate	2058.0	8.481083e-01	4.589093e-03	0.736430	8.479740e-01	8.480386e-01	8.481147e-01	1.000000e+00
_Continuous_Net_Profit_Growth_Rate	2058.0	2.173915e-01	5.678779e-03	0.000000	2.175741e-01	2.175961e-01	2.176198e-01	2.332046e-01
_Total_Asset_Growth_Rate	2058.0	5.287663e+09	2.912615e+09	0.000000	4.315000e+09	6.225000e+09	7.220000e+09	9.980000e+09
_Net_Value_Growth_Rate	2058.0	5.189504e+06	2.077918e+08	0.000000	4.362833e-04	4.554170e-04	4.883758e-04	9.330000e+09
_Total_Asset_Return_Growth_Rate_Ratio	2058.0	2.641004e-01	2.415661e-03	0.251620	2.637383e-01	2.640161e-01	2.643097e-01	3.586288e-01
_Cash_Reinvestment_perc	2058.0	3.771970e-01	2.737311e-02	0.025828	3.707295e-01	3.789678e-01	3.855575e-01	1.000000e+00
_Current_Ratio	2058.0	1.336249e+06	6.061917e+07	0.000000	6.567062e-03	8.945370e-03	1.350542e-02	2.750000e+09
_Quick_Ratio	2058.0	2.775510e+07	4.448654e+08	0.000000	2.946399e-03	5.284241e-03	8.902983e-03	9.230000e+09
_Interest_Expense_Ratio	2058.0	6.312913e-01	6.785512e-03	0.525126	6.306116e-01	6.307999e-01	6.317437e-01	8.121652e-01
_Total_debt_to_Total_net_worth	2037.0	1.071429e+07	2.696960e+08	0.000000	3.924894e-03	7.270721e-03	1.306869e-02	9.940000e+09
_Long_term_fund_suitability_ratio_A	2058.0	8.973310e-03	3.485186e-02	0.004129	5.162031e-03	5.517000e-03	6.415301e-03	1.000000e+00
_Net_profit_before_tax_to_Paid_in_capital	2058.0	1.753994e-01	2.622348e-02	0.000000	1.658623e-01	1.745683e-01	1.844450e-01	7.921047e-01
_Total_Asset_Turnover	2058.0	1.286405e-01	1.006216e-01	0.000000	6.146927e-02	1.034483e-01	1.679160e-01	9.190405e-01
_Accounts_Receivable_Turnover	2058.0	4.159864e+07	5.047673e+08	0.000000	7.446260e-04	1.081432e-03	1.854463e-03	9.740000e+09
_Average_Collection_Days	2058.0	2.629786e+07	4.109967e+08	0.000000	3.576384e-03	6.001272e-03	8.638997e-03	8.800000e+09
_Inventory_Turnover_Rate_times	2058.0	2.030227e+09	3.077250e+09	0.000000	1.909297e-04	1.910000e+07	3.815000e+09	9.990000e+09
_Fixed_Assets_Turnover_Frequency	2058.0	1.230898e+09	2.649289e+09	0.000000	2.278950e-04	5.995245e-04	8.423224e-03	9.990000e+09
_Net_Worth_Turnover_Rate_times	2058.0	3.957710e-02	4.239591e-02	0.008871	2.048387e-02	2.870968e-02	4.435484e-02	1.000000e+00
_Operating_profit_per_person	2058.0	4.036693e-01	5.358970e-02	0.000000	3.913864e-01	3.950781e-01	4.008927e-01	1.000000e+00
_Allocation_rate_per_person	2058.0	5.725559e+06	1.979500e+08	0.000000	4.671612e-03	1.062969e-02	2.457491e-02	8.280000e+09

_Quick_Assets_to_Total_Assets	2058.0	3.421979e-01	2.103925e-01	0.000000	1.734827e-01	3.061276e-01	4.845435e-01	9.8
_Cash_to_Total_Assets	1962.0	7.993675e-02	9.862260e-02	0.000184	2.061909e-02	4.563187e-02	9.771301e-02	9.250180e-01
_Quick_Assets_to_Current_Liability	2058.0	1.190476e+07	3.122923e+08	0.000000	3.616304e-03	5.972976e-03	9.608533e-03	8.820000e+09
_Cash_to_Current_Liability	2058.0	9.282507e+07	7.851899e+08	0.000101	1.085476e-03	2.684338e-03	7.540535e-03	9.170000e+09
_Operating_Funds_to_Liability	2058.0	3.482338e-01	3.840302e-02	0.026274	3.377032e-01	3.450257e-01	3.541402e-01	1.000000e+00
_Inventory_to_Working_Capital	2058.0	2.777491e-01	1.844394e-02	0.000000	2.770093e-01	2.772511e-01	2.777111e-01	1.000000e+00
_Inventory_to_Current_Liability	2058.0	5.786346e+07	6.278795e+08	0.000000	2.890842e-03	6.781166e-03	1.275116e-02	9.600000e+09
_Long_term_Liability_to_Current_Assets	2058.0	7.340107e+07	6.693526e+08	0.000000	0.000000e+00	2.587130e-03	1.049684e-02	9.310000e+09
_Retained_Earnings_to_Total_Assets	2058.0	9.303546e-01	2.976067e-02	0.000000	9.278868e-01	9.350756e-01	9.409371e-01	9.727688e-01
_Total_income_to_Total_expense	2058.0	2.357977e-03	4.644258e-04	0.000000	2.186964e-03	2.297452e-03	2.433146e-03	1.028413e-02
_Total_expense_to_Assets	2058.0	3.109208e-02	3.870042e-02	0.000853	1.270426e-02	2.086322e-02	3.530120e-02	1.000000e+00
_Current_Asset_Turnover_Rate	2058.0	1.273303e+09	2.839741e+09	0.000000	1.504698e-04	2.461660e-04	1.264005e-03	9.990000e+09
_Quick_Asset_Turnover_Rate	2058.0	2.571768e+09	3.453544e+09	0.000000	1.511758e-04	3.794085e-04	5.790000e+09	1.000000e+10
_Cash_Turnover_Rate	2058.0	2.653696e+09	2.821245e+09	0.000100	1.737418e-03	1.730000e+09	4.550000e+09	9.990000e+09
_Fixed_Assets_to_Assets	2058.0	4.042760e+06	1.834006e+08	0.000000	9.650577e-02	2.138107e-01	4.150287e-01	8.320000e+09
_Cash_Flow_to_Total_Assets	2058.0	6.442325e-01	4.505929e-02	0.000000	6.333645e-01	6.432462e-01	6.541577e-01	1.000000e+00
_Cash_Flow_to_Liability	2058.0	4.599747e-01	3.288112e-02	0.032583	4.574802e-01	4.593408e-01	4.617433e-01	9.051198e-01

_Fixed_Assets_to_Assets	2058.0	4.042760e+06	1.834006e+08	0.000000	9.650577e-02	2.138107e-01	4.150287e-01	8.32
_Cash_Flow_to_Total_Assets	2058.0	6.442325e-01	4.505929e-02	0.000000	6.333645e-01	6.432462e-01	6.541577e-01	1.000000e+00
_Cash_Flow_to_Liability	2058.0	4.599747e-01	3.288112e-02	0.032583	4.574802e-01	4.593408e-01	4.617433e-01	9.051198e-01
_CFO_to_Assets	2058.0	5.797344e-01	6.375060e-02	0.000000	5.503790e-01	5.825431e-01	6.123215e-01	9.751973e-01
_Cash_Flow_to_Equity	2058.0	3.146292e-01	1.277967e-02	0.000000	3.127830e-01	3.146423e-01	3.165460e-01	5.692307e-01
_Current_Liability_to_Current_Assets	2044.0	3.935178e-02	4.797815e-02	0.000000	2.177539e-02	3.265229e-02	4.394684e-02	1.000000e+00
_Liability_Assets_Flag	2058.0	3.401361e-03	5.823606e-02	0.000000	0.000000e+00	0.000000e+00	0.000000e+00	1.000000e+00
_Total_assets_to_GNP_price	2058.0	2.779397e+07	4.717714e+08	0.000000	9.124052e-04	2.479550e-03	7.004449e-03	9.820000e+09
_No_credit_Interval	2058.0	6.236856e-01	1.163052e-02	0.408682	6.233274e-01	6.237496e-01	6.240452e-01	9.563871e-01
_Degree_of_Financial_Leverage_DFL	2058.0	2.785248e-02	1.383854e-02	0.012845	2.677558e-02	2.681466e-02	2.702943e-02	4.643880e-01
_Interest_Coverage_Ratio_Interest_expense_to_EBIT	2058.0	5.654355e-01	1.153538e-02	0.172065	5.651580e-01	5.653149e-01	5.662324e-01	6.667613e-01
_Net_Income_Flag	2058.0	1.000000e+00	0.000000e+00	1.000000	1.000000e+00	1.000000e+00	1.000000e+00	1.000000e+00
_Equity_to_Liability	2058.0	4.252852e-02	5.952518e-02	0.003946	2.040787e-02	2.846004e-02	4.343255e-02	1.000000e+00
Default	2058.0	1.068999e-01	3.090610e-01	0.000000	0.000000e+00	0.000000e+00	0.000000e+00	1.000000e+00

- No duplicates found in the dataset also 298 values missing in the dataset.
- 0.2 % of data is missing out of total provided dataset

Treating the Outliers

The approach consists of calculating the outliers. After getting the visualizations and counts the outlies should be treated by applying trimming.

Checking for outliers by generation the boxplots:

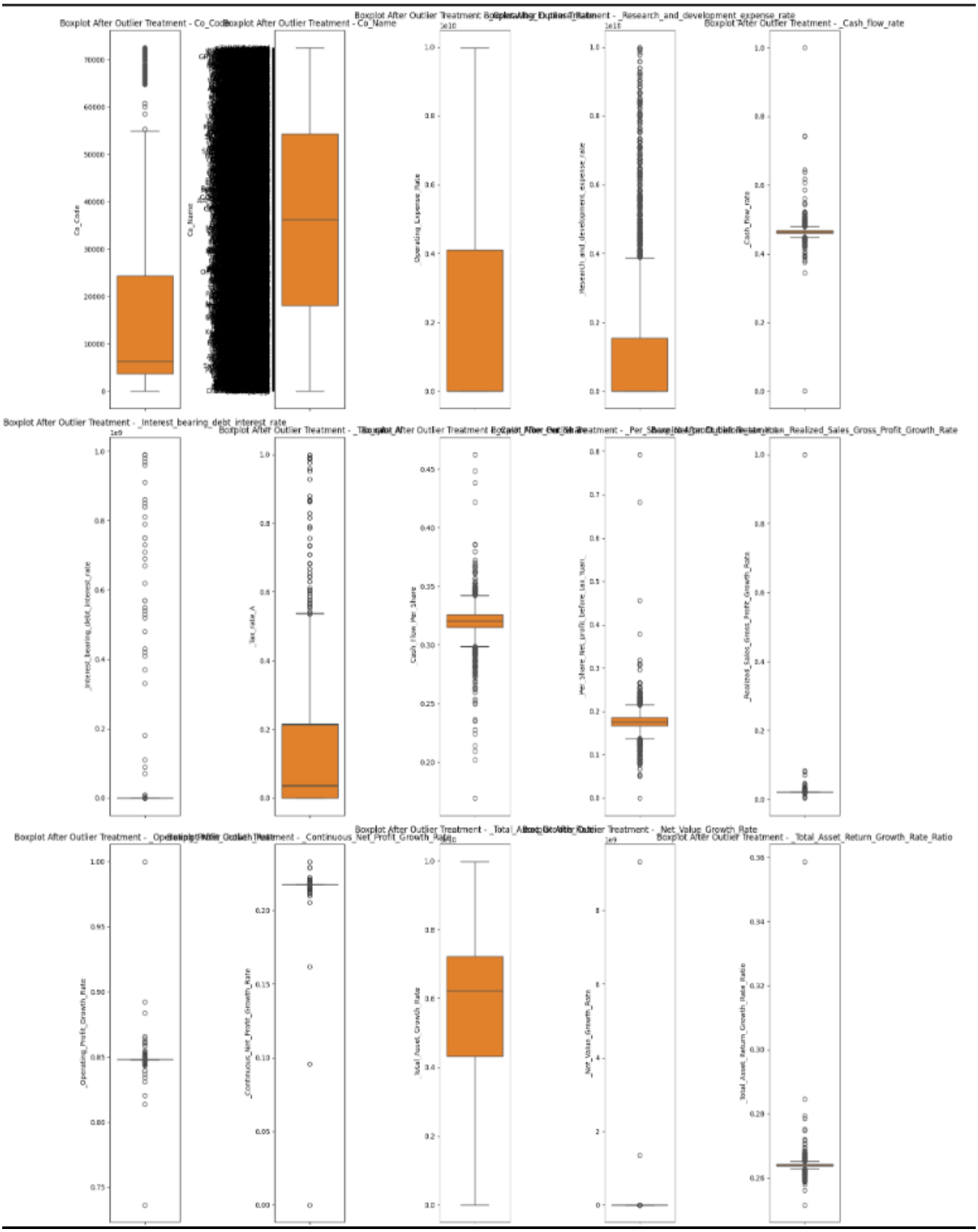


Figure 1: Outlier boxplots – Before and after treating variables - 1

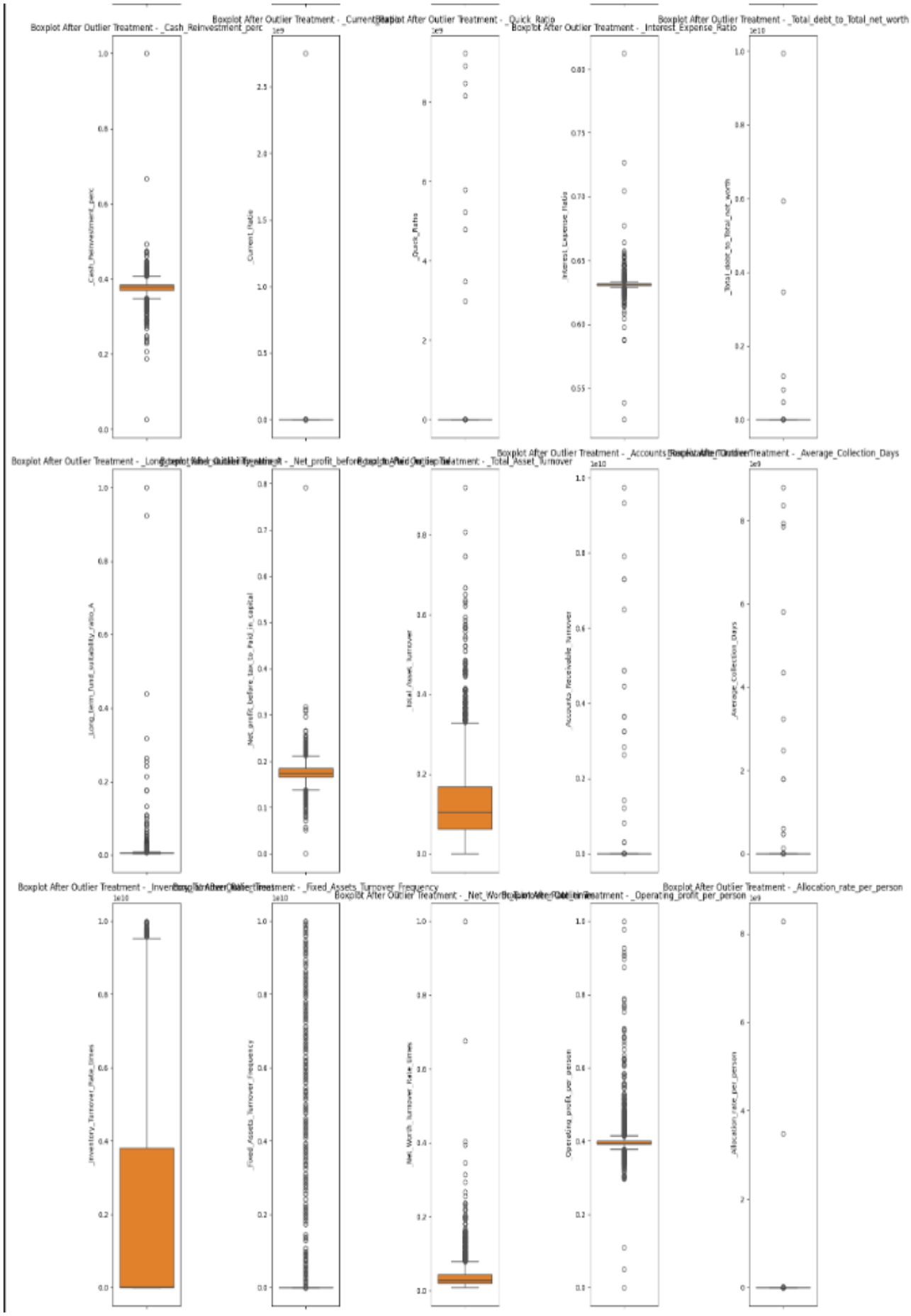


Figure 2: Outlier boxplots – Before and after treating variables – 2

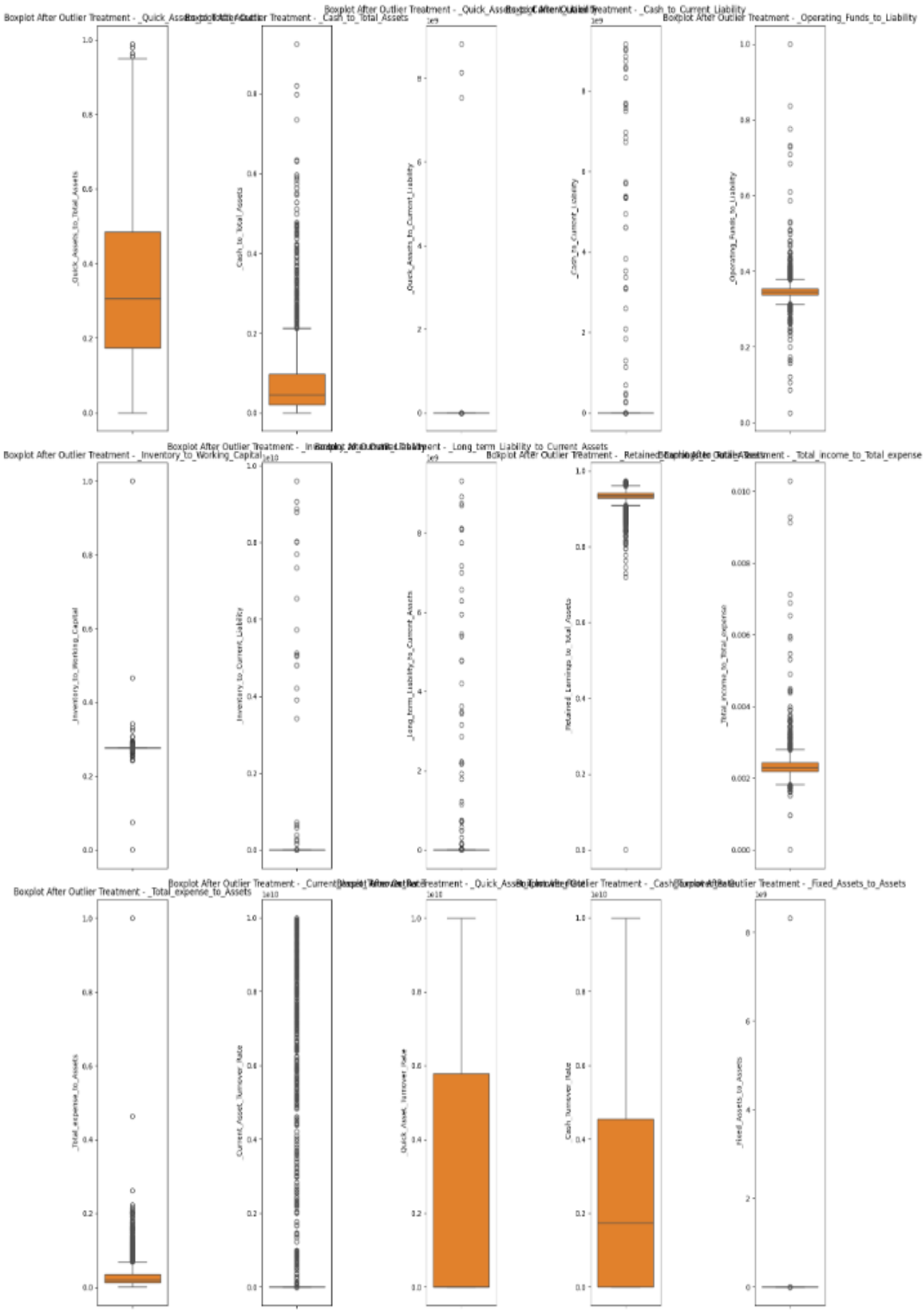


Figure 3: Outlier boxplots – Before and after treating variables – 3

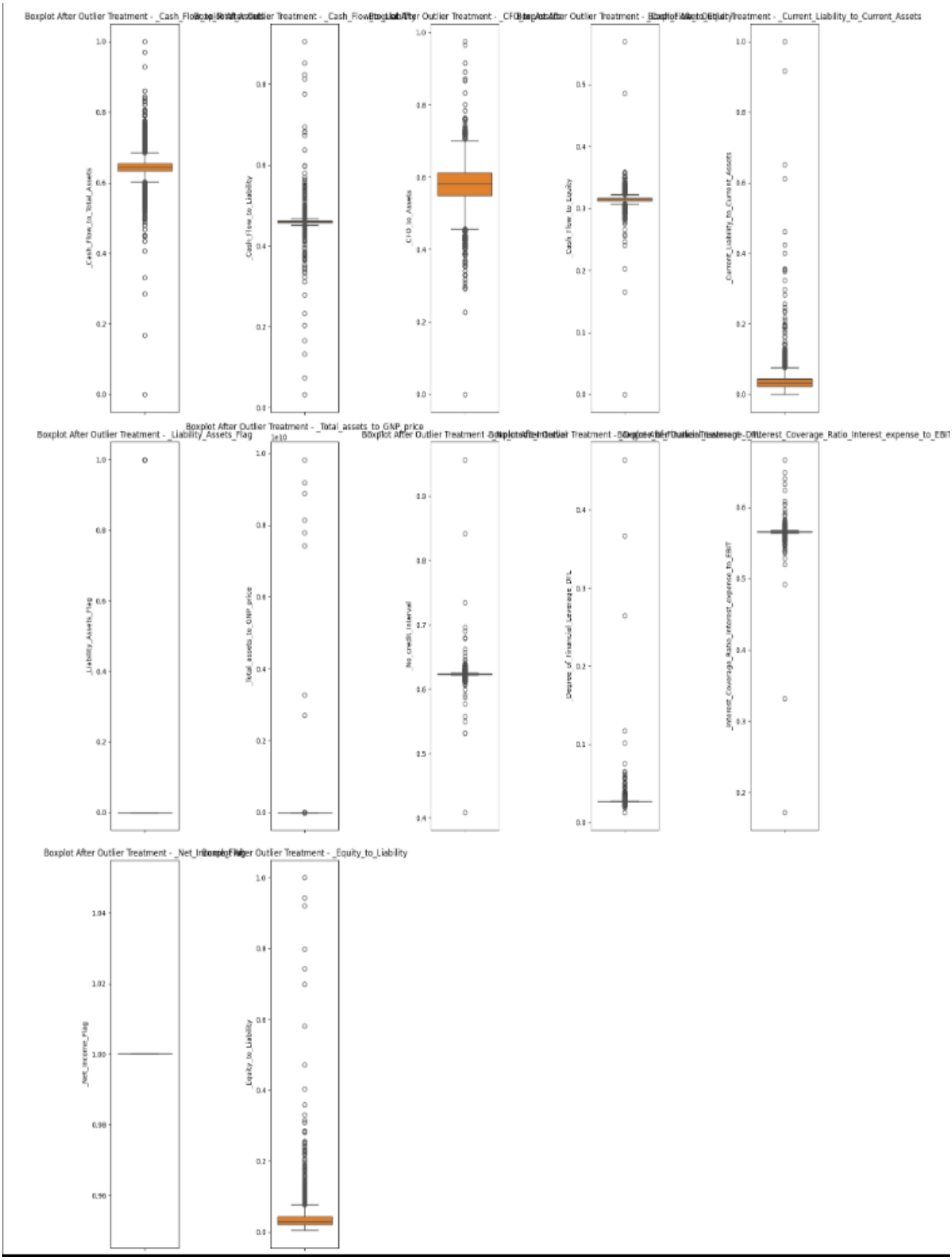


Figure 4: Outlier boxplots – Before and after treating variables – 4

```
Outliers Before Treatment:
Co_Code                                                    0
Co_Name                                                    0
_Accounts_Receivable_Turnover                             52
_Allocation_rate_per_person                               25
_Average_Collection_Days                                  39
_CFO_to_Assets                                             3
_Cash_Flow_Per_Share                                      12
_Cash_Flow_to_Equity                                      14
_Cash_Flow_to_Liability                                   54
_Cash_Flow_to_Total_Assets                                 7
_Cash_Reinvestment_perc                                    9
_Cash_Turnover_Rate                                        0
_Cash_flow_rate                                           22
_Cash_to_Current_Liability                                52
_Cash_to_Total_Assets                                      4
_Continuous_Net_Profit_Growth_Rate                        68
_Current_Asset_Turnover_Rate                               0
_Current_Liability_to_Current_Assets                      23
_Current_Ratio                                            19
_Degree_of_Financial_Leverage_DFL                         62
_Equity_to_Liability                                      18
_Fixed_Assets_Turnover_Frequency                           0
_Fixed_Assets_to_Assets                                    1
_Interest_Coverage_Ratio_Interest_expense_to_EBIT         34
_Interest_Expense_Ratio                                   46
_Interest_bearing_debt_interest_rate                      68
_Inventory_Turnover_Rate_times                             0
_Inventory_to_Current_Liability                           42
_Inventory_to_Working_Capital                             49
_Liability_Assets_Flag                                     7
_Long_term_Liability_to_Current_Assets                    52
_Long_term_fund_suitability_ratio_A                       46
_Net_Income_Flag                                           0
_Net_Value_Growth_Rate                                    31
_Net_Worth_Turnover_Rate_times                            13
_Net_profit_before_tax_to_Paid_in_capital                  3
_No_credit_Interval                                       42
_Operating_Expense_Rate                                    0
_Operating_Funds_to_Liability                             24
_Operating_Profit_Growth_Rate                             50
_Operating_profit_per_person                              35
_Per_Share_Net_profit_before_tax_Yuan_                     5
_Quick_Asset_Turnover_Rate                                 0
_Quick_Assets_to_Current_Liability                        25
_Quick_Assets_to_Total_Assets                              0
_Quick_Ratio                                              30
_Realized_Sales_Gross_Profit_Growth_Rate                  66
_Research_and_development_expense_rate                     0
_Retained_Earnings_to_Total_Assets                         8
_Tax_rate_A                                               14
_Total_Asset_Growth_Rate                                   0
_Total_Asset_Return_Growth_Rate_Ratio                     21
_Total_Asset_Turnover                                      2
_Total_assets_to_GNP_price                                35
_Total_debt_to_Total_net_worth                            26
_Total_expense_to_Assets                                   4
_Total_income_to_Total_expense                            16
dtype: int64
```

In the above output its has been observed that there are outliers in the given dataset
After treating the outliers, the output is as mentioned below

```
Outliers After Treatment:
Co_Code                                                    0
Co_Name                                                    0
_Accounts_Receivable_Turnover                              0
_Allocation_rate_per_person                                0
_Average_Collection_Days                                   0
_CFO_to_Assets                                             0
_Cash_Flow_Per_Share                                       0
_Cash_Flow_to_Equity                                       0
_Cash_Flow_to_Liability                                    0
_Cash_Flow_to_Total_Assets                                 0
_Cash_Reinvestment_perc                                    0
_Cash_Turnover_Rate                                        0
_Cash_flow_rate                                            0
_Cash_to_Current_Liability                                 0
_Cash_to_Total_Assets                                      0
_Continuous_Net_Profit_Growth_Rate                         0
_Current_Asset_Turnover_Rate                               0
_Current_Liability_to_Current_Assets                       0
_Current_Ratio                                             0
_Degree_of_Financial_Leverage_DFL                          0
_Equity_to_Liability                                       0
_Fixed_Assets_Turnover_Frequency                           0
_Fixed_Assets_to_Assets                                    0
_Interest_Coverage_Ratio_Interest_expense_to_EBIT          0
_Interest_Expense_Ratio                                    0
_Interest_bearing_debt_interest_rate                       0
_Inventory_Turnover_Rate_times                             0
_Inventory_to_Current_Liability                            0
_Inventory_to_Working_Capital                              0
_Liability_Assets_Flag                                     0

_Long_term_Liability_to_Current_Assets                     0
_Long_term_fund_suitability_ratio_A                        0
_Net_Income_Flag                                           0
_Net_Value_Growth_Rate                                     0
_Net_Worth_Turnover_Rate_times                             0
_Net_profit_before_tax_to_Paid_in_capital                  0
_No_credit_Interval                                        0
_Operating_Expense_Rate                                    0
_Operating_Funds_to_Liability                              0
_Operating_Profit_Growth_Rate                              0
_Operating_profit_per_person                               0
_Per_Share_Net_profit_before_tax_Yuan_                     0
_Quick_Asset_Turnover_Rate                                 0
_Quick_Assets_to_Current_Liability                         0
_Quick_Assets_to_Total_Assets                              0
_Quick_Ratio                                               0
_Realized_Sales_Gross_Profit_Growth_Rate                   0
_Research_and_development_expense_rate                     0
_Retained_Earnings_to_Total_Assets                         0
_Tax_rate_A                                                0
_Total_Asset_Growth_Rate                                   0
_Total_Asset_Return_Growth_Rate_Ratio                      0
_Total_Asset_Turnover                                      0
_Total_assets_to_GNP_price                                 0
_Total_debt_to_Total_net_worth                             0
_Total_expense_to_Assets                                   0
_Total_income_to_Total_expense                             0
dtype: int64
```

2. Treating the missing values

- 298 null values found in the dataset

In the below heatmap it has been observed that most of the missing values are present in the 2 variables are cash flow per share and cash to total assets.

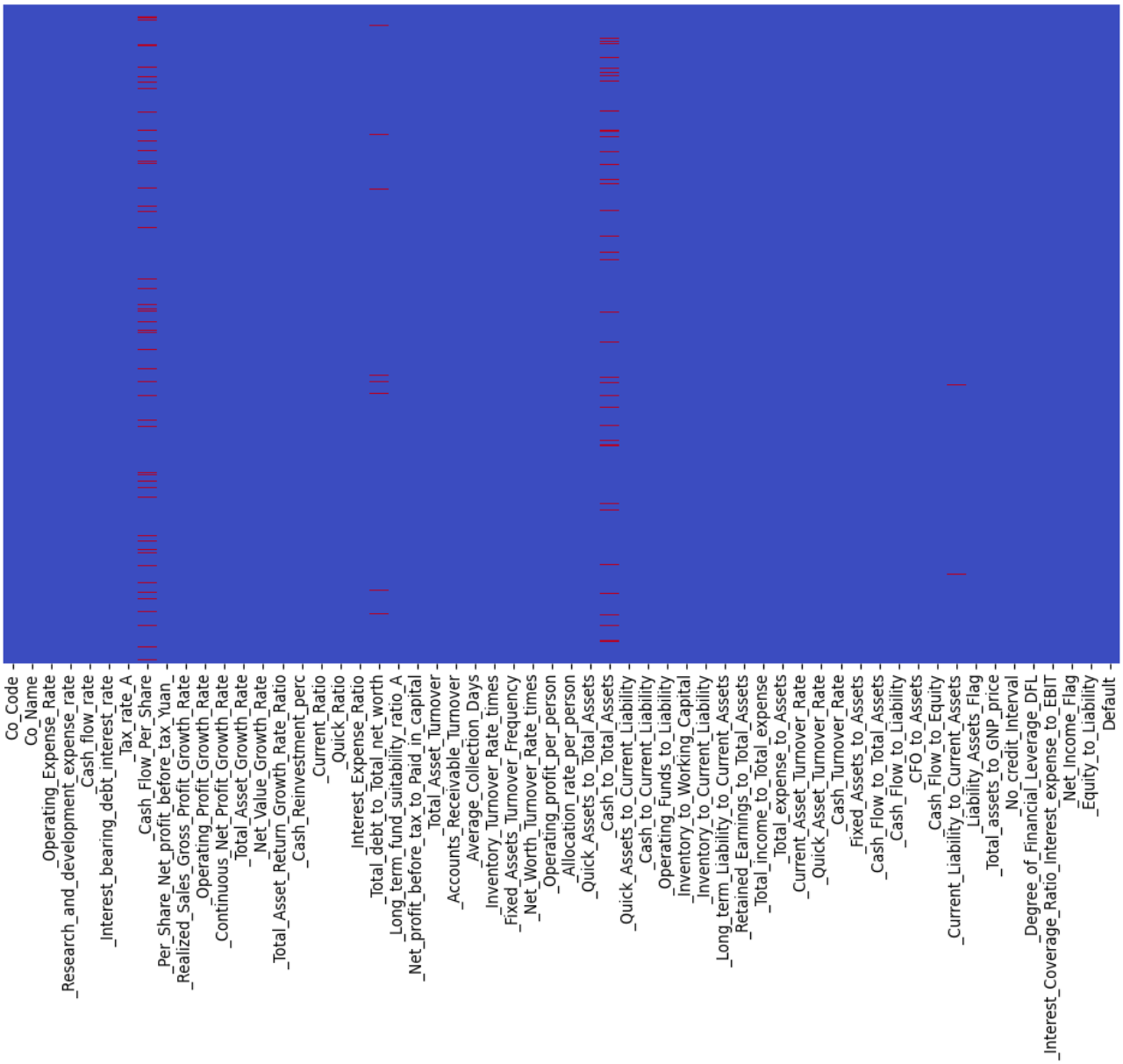


Figure 5: Heatmap representing null values in the dataset

List of variables which have missing values in the dataset are

```
_Cash_Flow_Per_Share                    167
_Total_debt_to_Total_net_worth           21
_Cash_to_Total_Assets                    96
_Current_Liability_to_Current_Assets     14
dtype: int64
```

Calculating the mean of the missing variables and replacing the null values with mean value

Output:

```
Output after replacing the null values with mean values Co_Code
Co_Name                                                  0
_Operating_Expense_Rate                                  0
_Research_and_development_expense_rate                   0
_Cash_flow_rate                                          0
_Interest_bearing_debt_interest_rate                     0
_Tax_rate_A                                              0
_Cash_Flow_Per_Share                                     0
_Per_Share_Net_profit_before_tax_Yuan_                   0
_Realized_Sales_Gross_Profit_Growth_Rate                 0
_Operating_Profit_Growth_Rate                            0
_Continuous_Net_Profit_Growth_Rate                       0
_Total_Asset_Growth_Rate                                 0
_Net_Value_Growth_Rate                                   0
_Total_Asset_Return_Growth_Rate_Ratio                    0
_Cash_Reinvestment_perc                                  0
_Current_Ratio                                           0
_Quick_Ratio                                             0
_Interest_Expense_Ratio                                  0
_Total_debt_to_Total_net_worth                           0
_Long_term_fund_suitability_ratio_A                      0
_Net_profit_before_tax_to_Paid_in_capital                0
_Total_Asset_Turnover                                    0
_Accounts_Receivable_Turnover                            0
_Average_Collection_Days                                 0
_Inventory_Turnover_Rate_times                           0
```

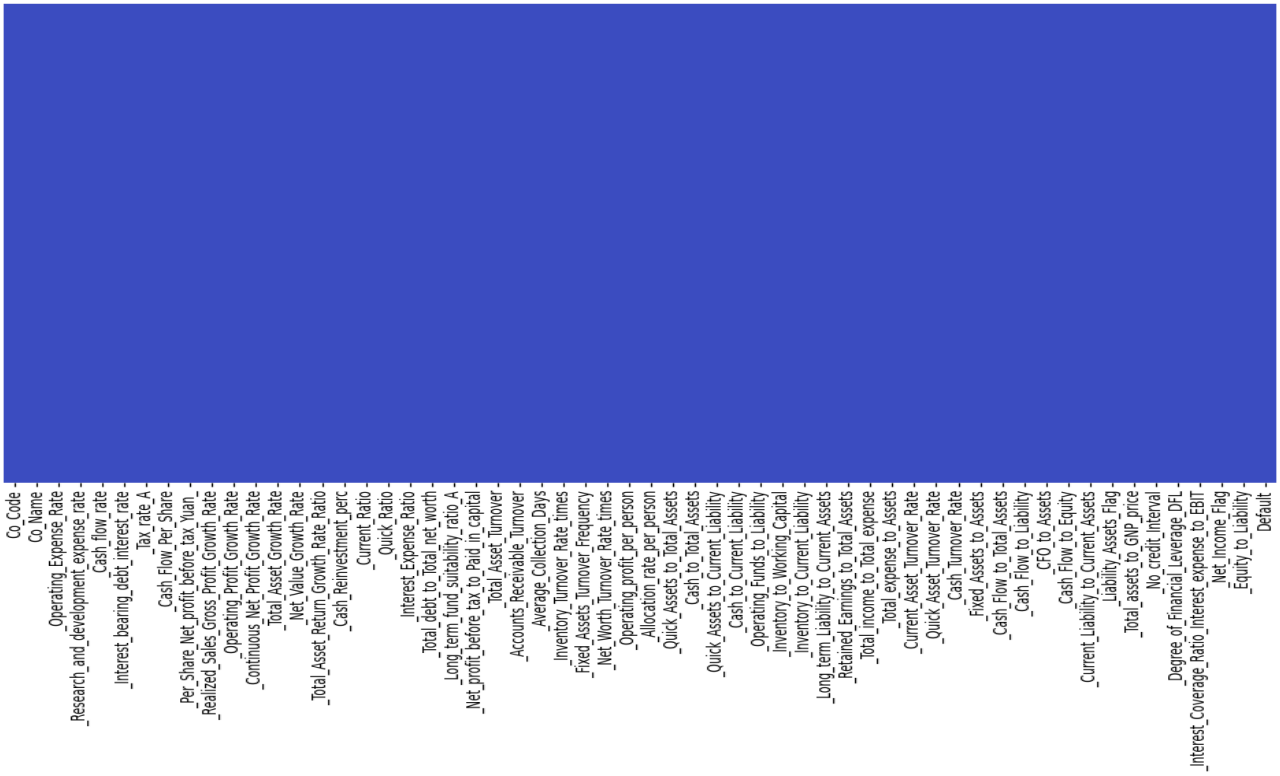


Figure 6: Heatmap representing no null values after treating the data

3. Univariate & Bivariate analysis with proper interpretation. (You may choose to include only those variables which were significant in the model building)

Univariate analysis

1. Default Variable

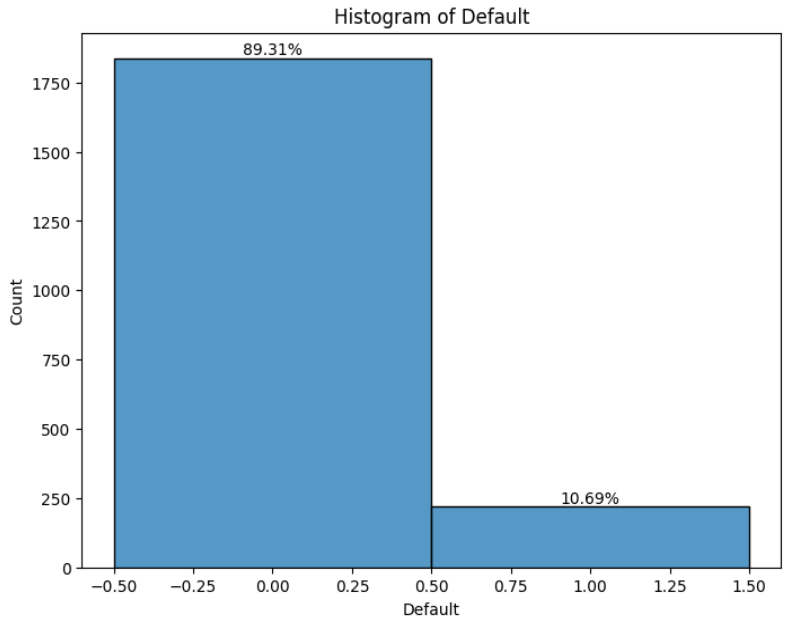


Figure 7: Histogram for Default and Not Defaulted
As per the above histogram it is observed 89% of data is Not -Defaulted and 10.6% of data is Defaulted – so 220 went bankrupted.

2. Operating Expense rate variable

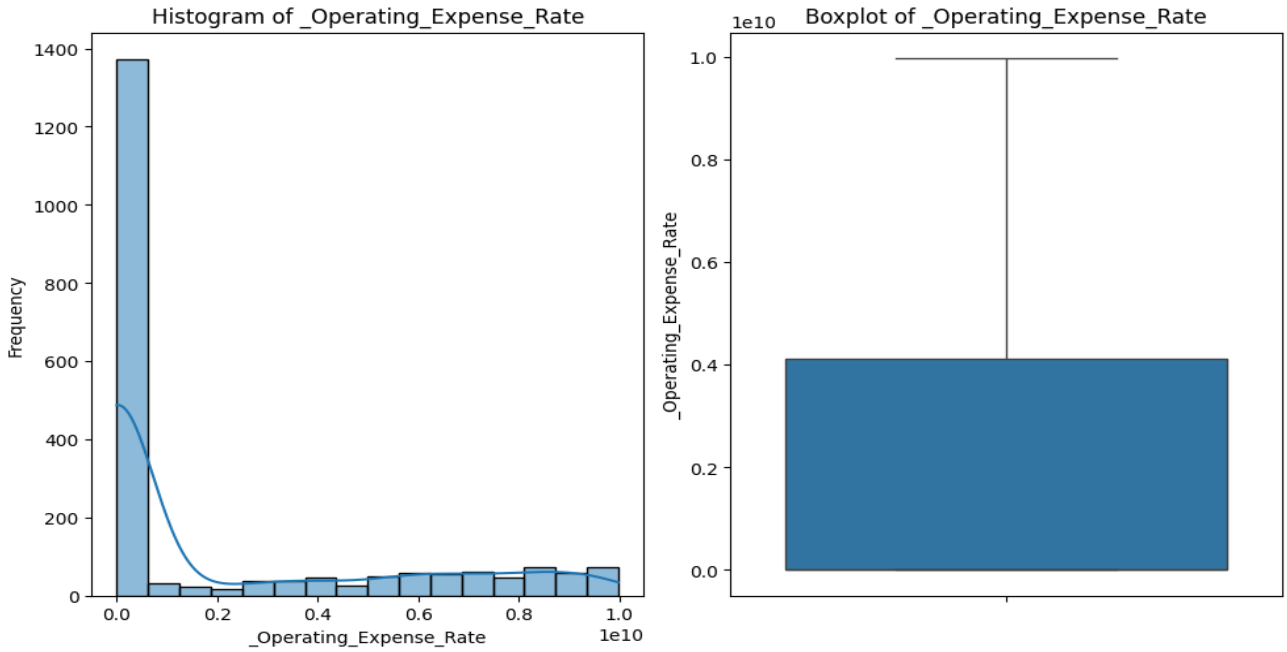


Figure 8: Histogram for operating Expense rate
Most of the data lies in the 0 value and distributed till the 1 and the boxplot refers the same

3. Research and Expense development rate variable

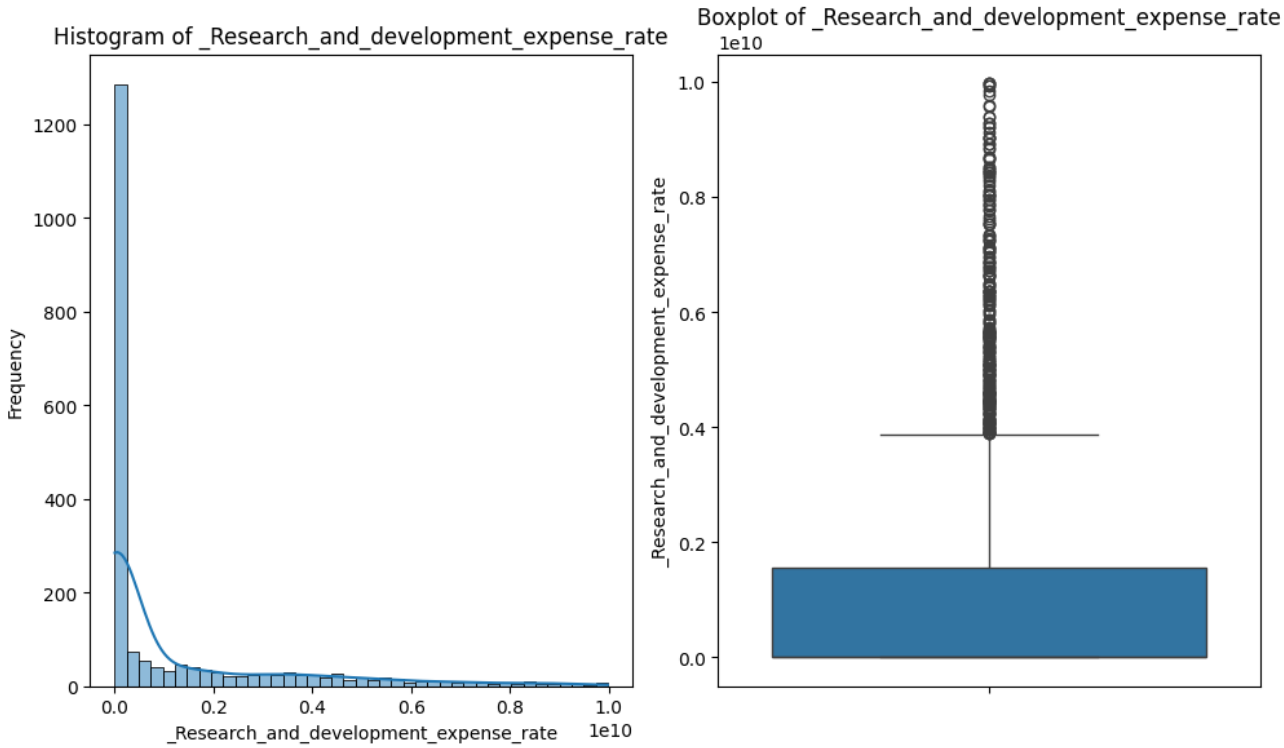


Figure 9: Histogram for Research and Expense development rate
In the above Boxplot it has been observed that data lies in the upper limit in between 0.4 to 1

4. Research and Expense development rate variable

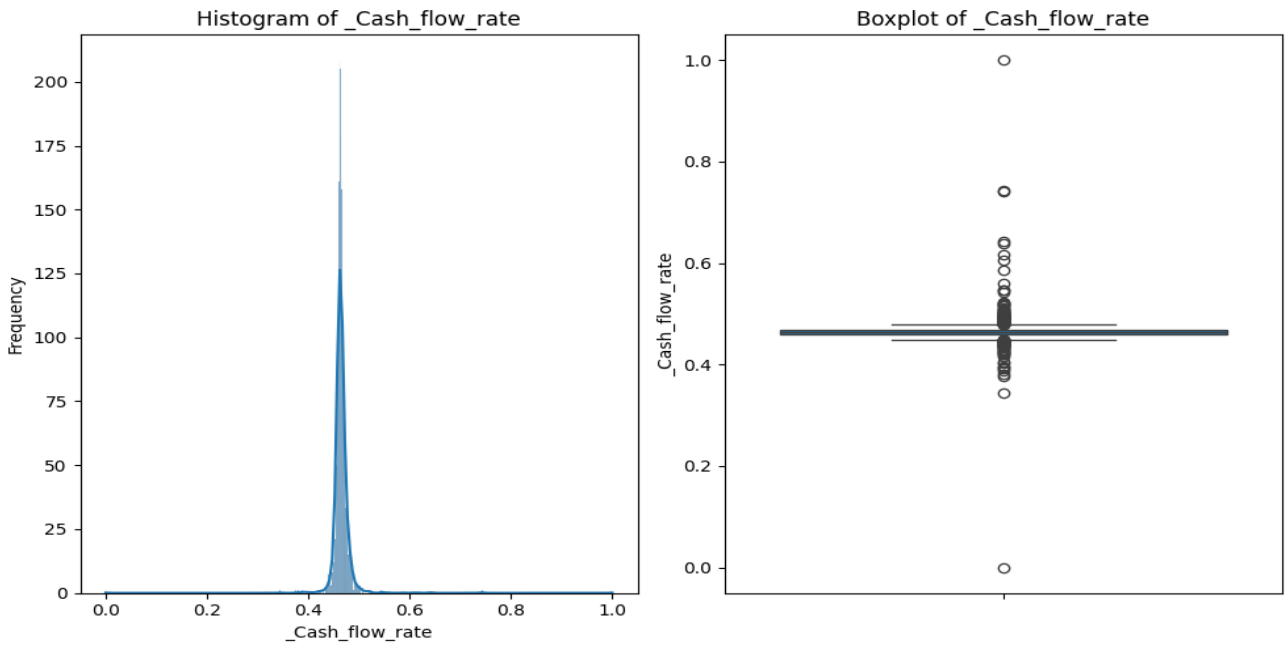


Figure 10: Histogram for Cash flow rate

In the above graph it has been observed that most of the data lies in between 0.4 to 0.6 range

Bivariate analysis

1. Correlation heatmap for all the variables

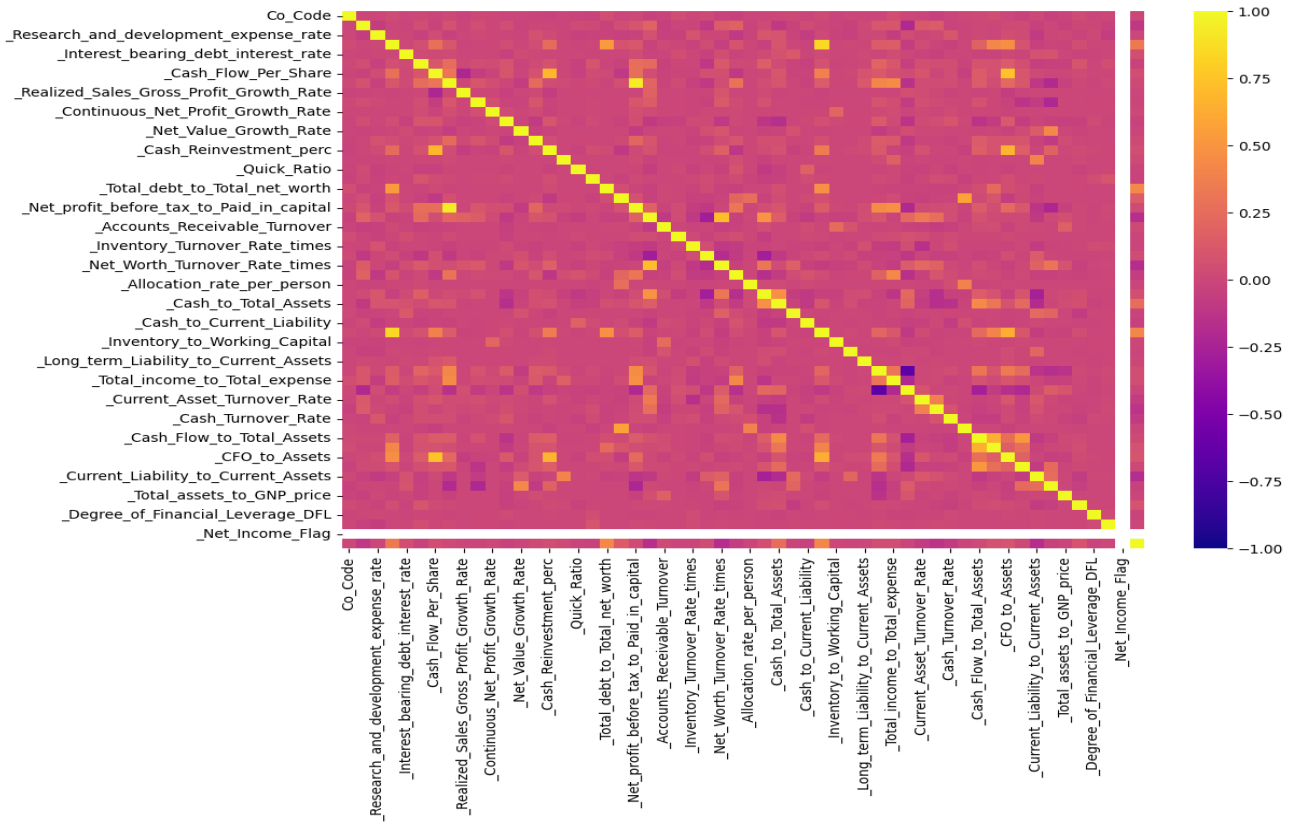


Figure 11: Correlation heatmap for all the variables

2. Scatter plot between 'Operating Expense Rate' and 'Cash flow rate'

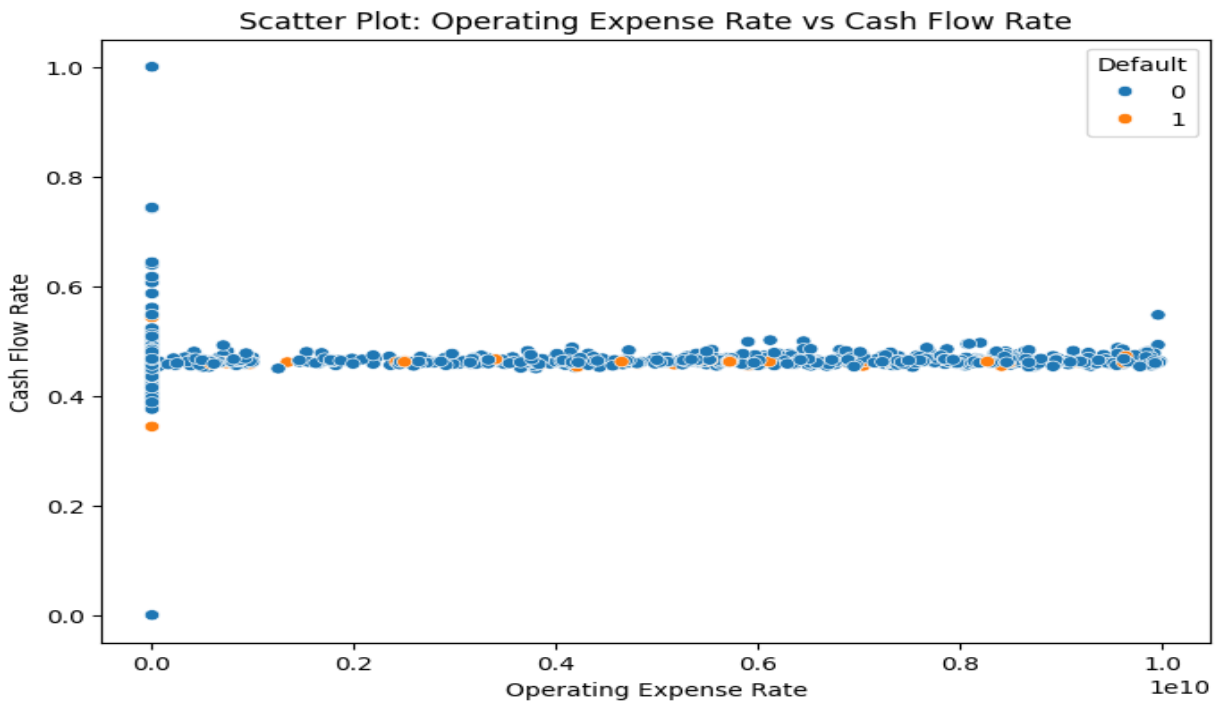


Figure 12: Scatter plot - Operating Expense Rate and Cash flow rate

3. **Box plot for 'Operating Expense Rate' by 'Default'**

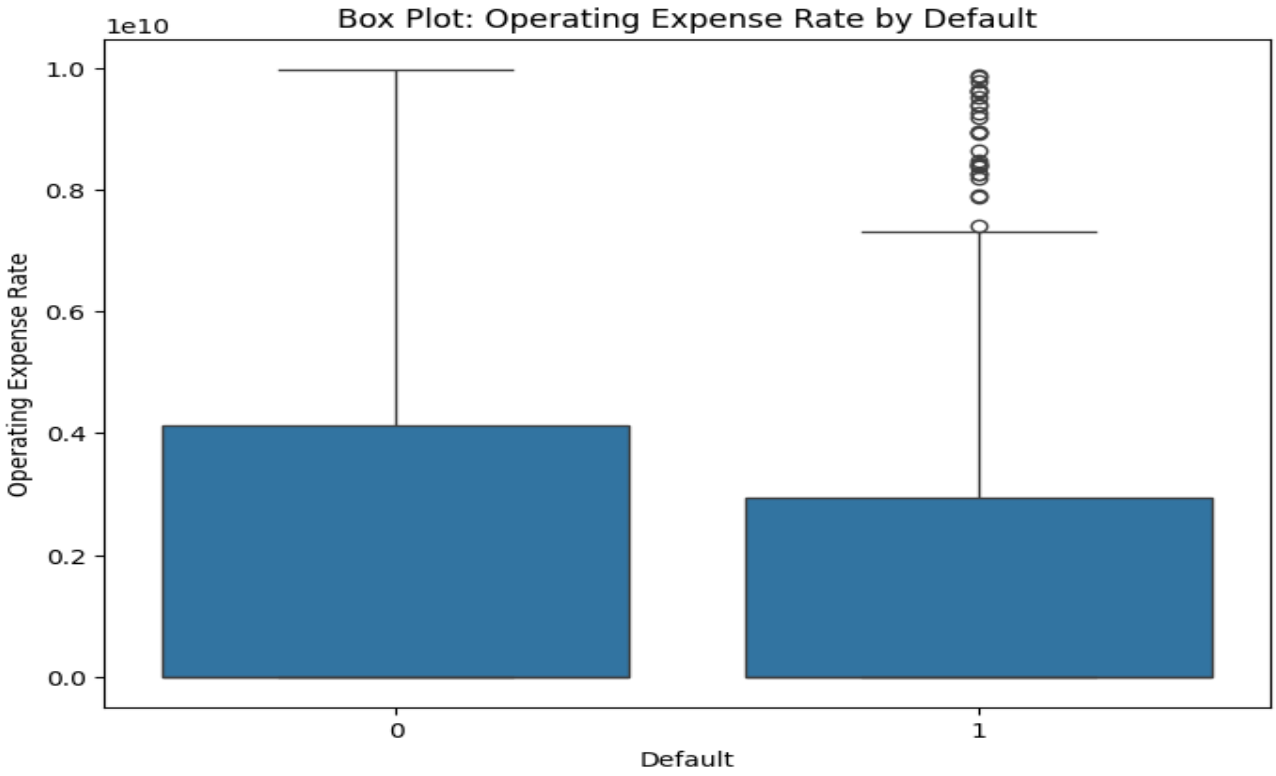


Figure 13: Box plot for 'Operating Expense Rate' by 'Default'

We can observe that defaulted have more operating expense rate

Histogram of Cash Flow Per Share by Default

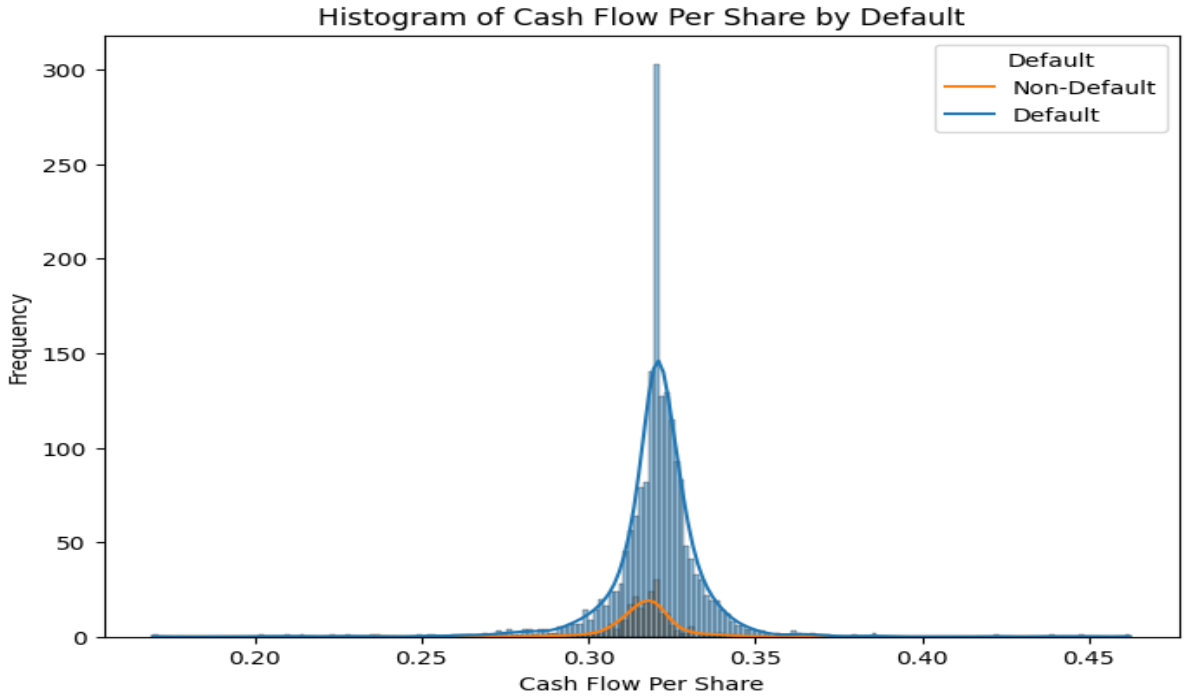


Figure 14: Histogram of Cash Flow Per Share by Default

5. **Bar plot for 'Interest Coverage Ratio Interest expense to EBIT' by 'Default'**

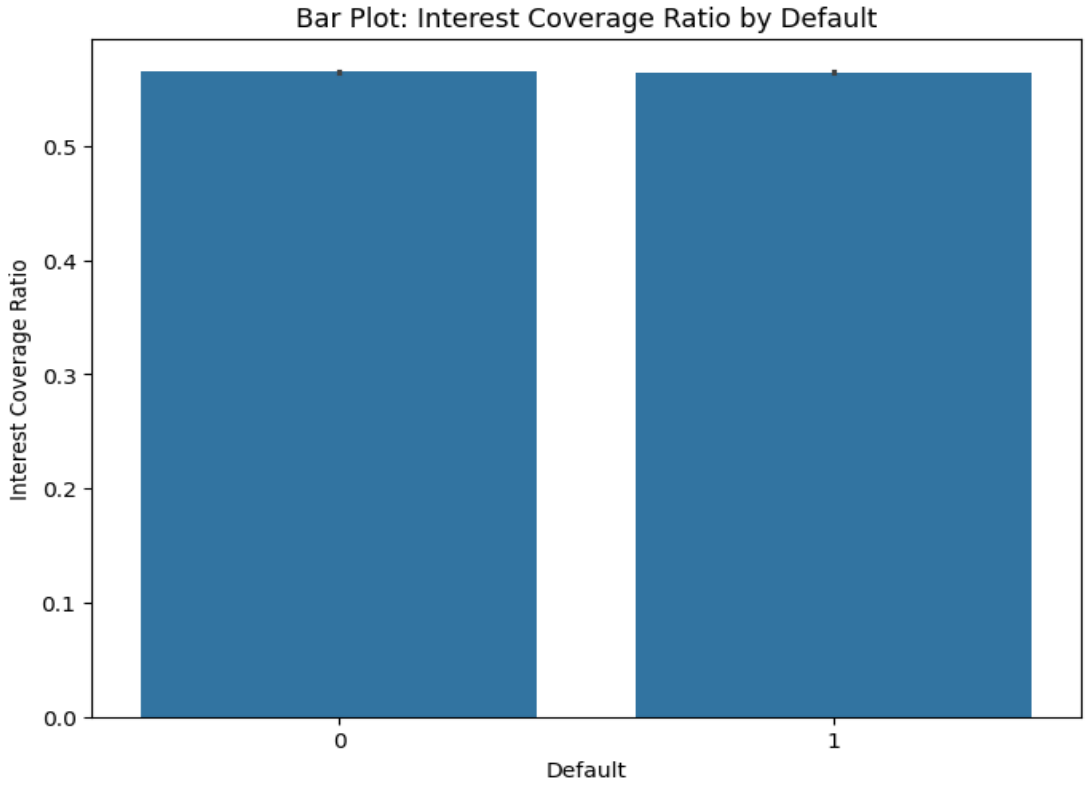


Figure 15: Bar plot for 'Interest Coverage Ratio Interest expense to EBIT' by 'Default'

6. **Violin plot for 'Tax rate A' by 'Default'**

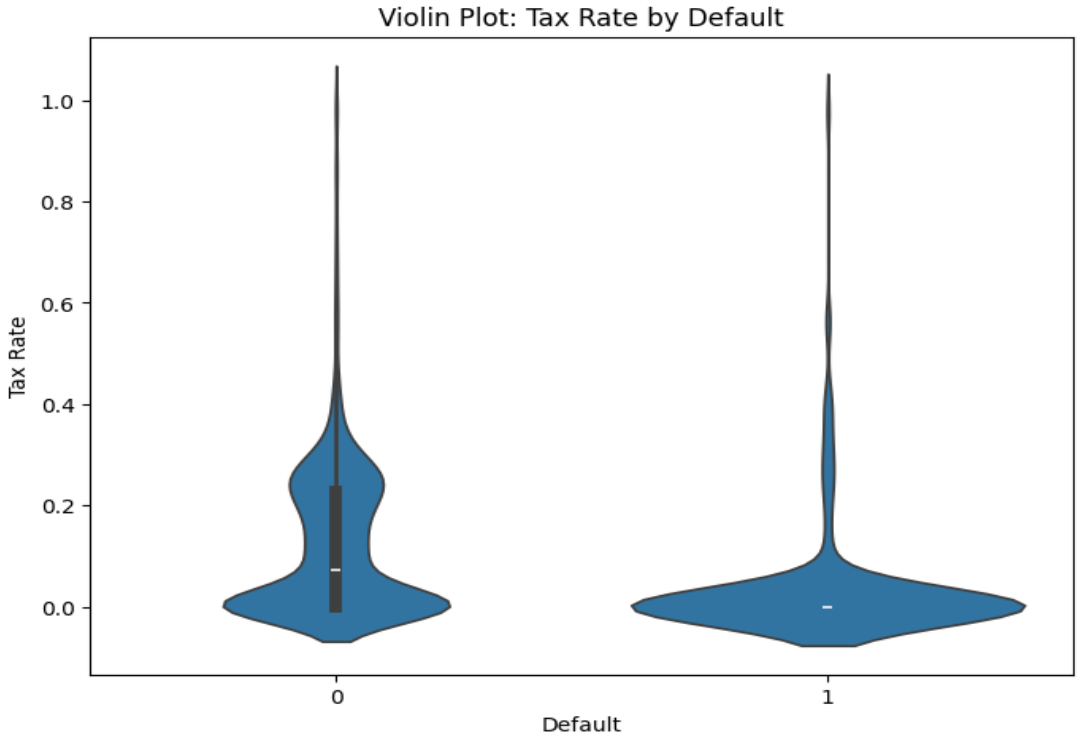


Figure 16: Violin plot for 'Tax rate A' by 'Default'

It has been observed that the bankrupted companies not paid even tax rates.

4. Train Test Split

As mentioned in the problem statement the data is spitted into train and test data basis of Split the data into train and test datasets in the ratio of 67:33 and use a random state of 42 (*random_state=42*). Model building is to be done on the train dataset and model validation is to be done on the test dataset.

```
Count of train data:
1378

Count of test data:
680
```

5. Build Logistic Regression Model (using statsmodels library) on most important variables on train dataset and choose the optimum cut-off. Also showcase your model building approach

- 4 important variables are selected to build the logistic regression model

Output of the model:

```
Optimization terminated successfully.
         Current function value: 0.318781
         Iterations 8
                           Logit Regression Results
==============================================================================
Dep. Variable:                Default   No. Observations:                 1378
Model:                          Logit   Df Residuals:                     1372
Method:                           MLE   Df Model:                            5
Date:                Sun, 25 Feb 2024   Pseudo R-squ.:                  0.08571
Time:                        14:17:00   Log-Likelihood:                -439.28
converged:                       True   LL-Null:                       -480.46
Covariance Type:            nonrobust   LLR p-value:                 2.691e-16
==========================================================================================
                                            coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------------------
const                                     4.1692      2.230      1.869      0.062      -0.202       8.540
_Operating_Expense_Rate              -4.903e-12   2.73e-11     -0.180      0.858   -5.84e-11    4.86e-11
_Cash_flow_rate                          -4.1006      3.742     -1.096      0.273     -11.434       3.233
_Interest_bearing_debt_interest_rate -3.321e-09   3.27e-09     -1.016      0.310   -9.73e-09    3.09e-09
_Tax_rate_A                              -7.5397      1.181     -6.383      0.000      -9.855      -5.225
_Cash_Flow_Per_Share                    -12.0293      5.901     -2.039      0.041     -23.594      -0.464
==========================================================================================
```

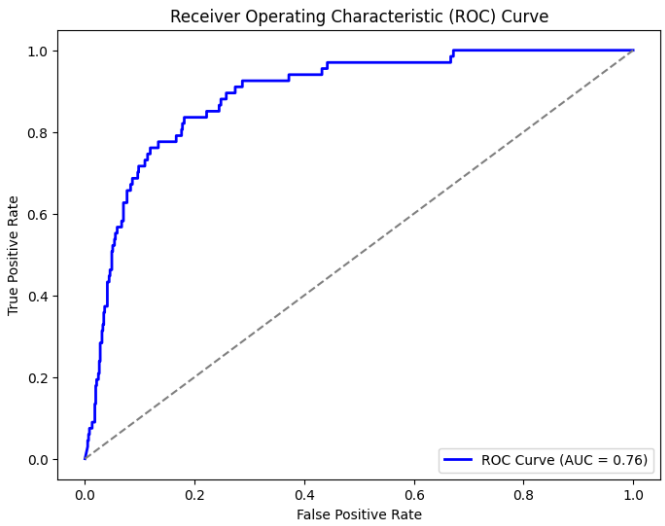
- No. Observations: 1378
- Pseudo R-squared: 0.08571
- The coefficient represents the change in the log-odds of the dependent variable (Default) for a one-unit change in the predictor variable, holding all other variables constant.
- For example, a one-unit increase in 'Tax rate A' results in a decrease of 7.5397 in the log-odds of 'Default'.
- The p-values associated with each coefficient test the null hypothesis that the coefficient is equal to zero (no effect).
- Lower p-values indicate greater evidence against the null hypothesis and suggest that the predictor variable is statistically significant.
- In this output, 'Tax rate A' and 'Cash Flow Per Share' have p-values less than 0.05, indicating they are statistically significant at the 5% level.
- For example, the coefficient for 'Tax rate A' is -7.5397, suggesting that a one-unit increase in 'Tax rate A' is associated with a decrease of 7.5397 in the log-odds of 'Default'.

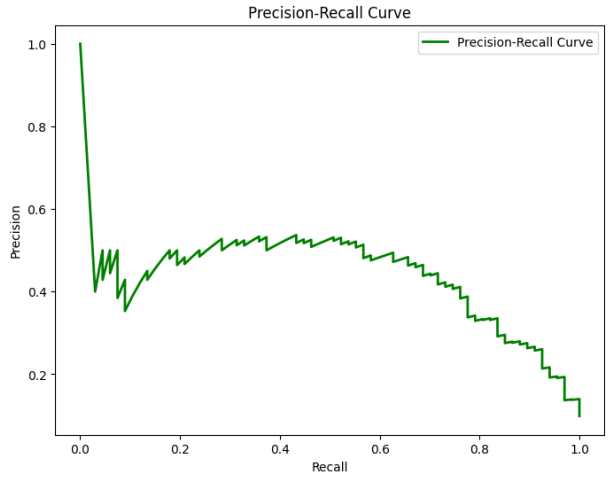
- For instance, the 95% confidence interval for 'Cash Flow Per Share' ranges from -23.594 to -0.464.

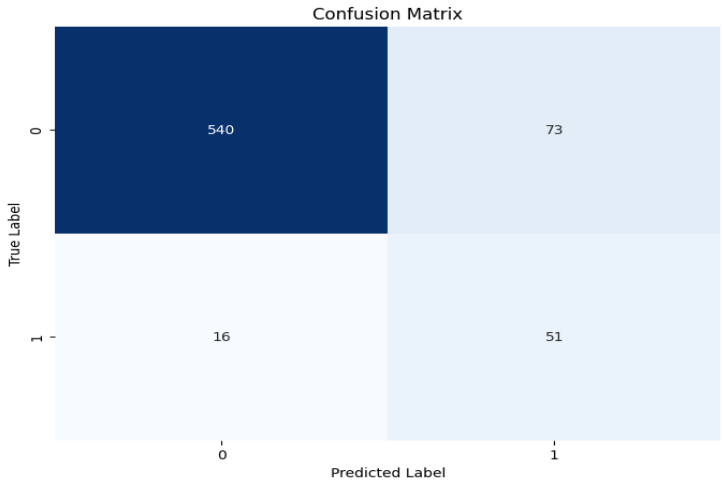
6. Validate the Model on Test Dataset and state the performance metrics. Also state interpretation from the model

Output:

```
AUC Score: 0.7552774463733535
Optimum Cutoff: 0.15749238747749514
```








```
Current function value: 0.189433
Iterations: 35
Performance Metrics on Test Dataset:
Accuracy: 0.9058823529411765
Precision: 0.5283018867924528
Recall: 0.417910447761194
F1-Score: 0.4666666666666667
ROC-AUC Score: 0.8919432202770812
```

Figure 17: Roc curve, Recall Curve, Confusion matrix and evaluation output - logistic regression model

- The logistic regression model was evaluated on the test dataset.
- It achieved a reasonably low current function value of 0.189433, indicating good model fit.
- The model demonstrated strong performance on the test dataset with an accuracy of 90.59%.
- Precision, which measures the proportion of correctly predicted positive cases out of all predicted positives, stood at 52.83%.
- Recall, which measures the proportion of correctly predicted positive cases out of all actual positives, was 41.79%.

- The F1-score, which is the harmonic mean of precision and recall, was 46.67%, indicating a balance between precision and recall.
- he ROC-AUC score, representing the area under the receiver operating characteristic curve, was 0.8919, indicating good overall model performance in distinguishing between positive and negative cases.
- The high accuracy and ROC-AUC score suggest that the logistic regression model is effective in classifying instances correctly.
- However, the relatively lower precision and recall values indicate that there may be room for improvement, especially in correctly identifying positive cases while minimizing false positives

7. Build a Random Forest Model on Train Dataset. Also showcase your model building approach

Output of the Model:

```
Confusion Matrix:
[[599  14]
 [ 40  27]]

Classification Report:
              precision    recall  f1-score   support

           0       0.94      0.98      0.96       613
           1       0.66      0.40      0.50        67

    accuracy                           0.92       680
   macro avg       0.80      0.69      0.73       680
weighted avg       0.91      0.92      0.91       680
```

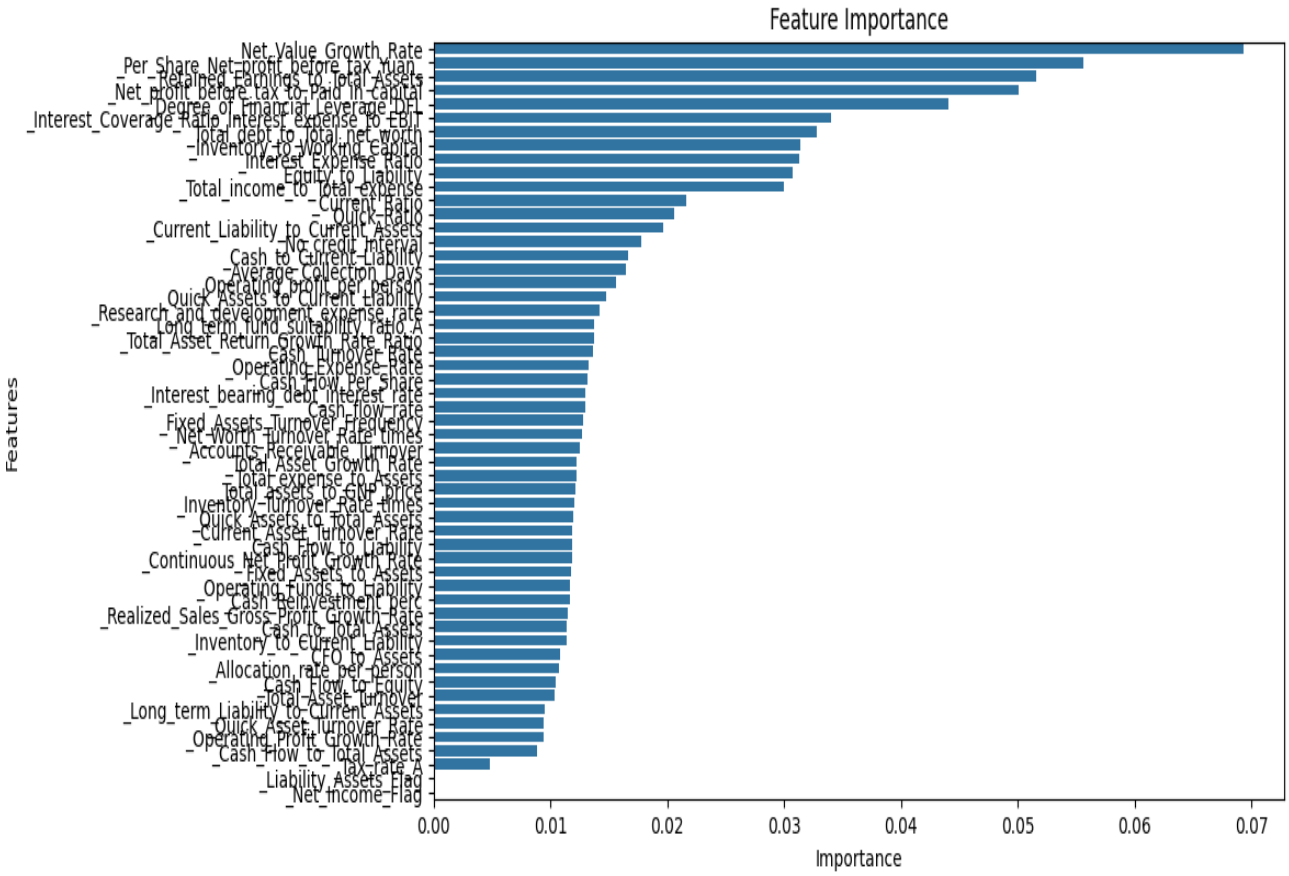


Figure 18: Feature importance for each variable

- The confusion matrix displays the model's performance in categorizing instances into true negatives, false positives, false negatives, and true positives.
- In this case, there are 599 true negatives, 27 true positives, 14 false positives, and 40 false negatives.
- Precision, recall, and F1-score are key metrics provided in the classification report, offering insights into the model's performance for each class (0 and 1).
- The report shows a precision of 0.94 for class 0, 0.66 for class 1, and a recall of 0.40 for class 1.
- The model's accuracy, representing the proportion of correctly classified instances, is reported as
- The model demonstrates strong performance in correctly classifying negative cases (class 0), with high precision (0.94).
- However, it exhibits lower precision (0.66) and recall (0.40) for positive cases (class 1), indicating potential challenges in correctly identifying positive instances.
- Further analysis is recommended to investigate factors contributing to misclassifications, particularly false negatives.

8. Validate the Random Forest Model on test Dataset and state the performance metrics. Also state interpretation from the model

Model evaluation o/p

```
Performance Metrics on Test Dataset:
Accuracy: 0.9205882352941176
Precision: 0.6585365853658537
Recall: 0.40298507462686567
F1-Score: 0.5
ROC-AUC Score: 0.9294270896739791
```

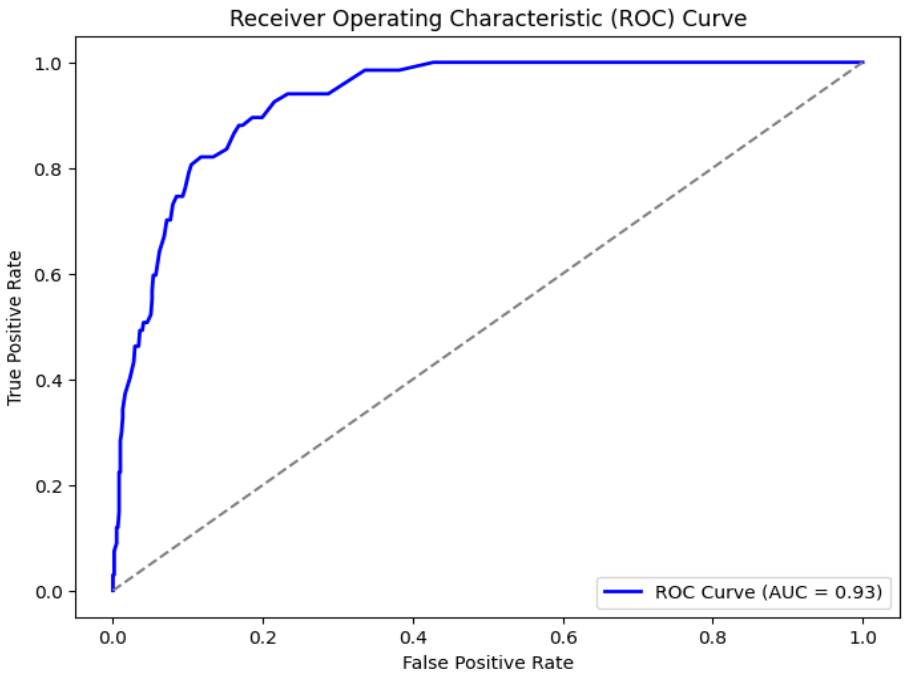


Figure 19: ROC Curve for Random Forest Model

- The accuracy of the model on the test dataset is 92.06%, indicating the proportion of correctly classified instances among all instances.
- The precision of the model is 65.85%, representing the accuracy of positive predictions. It indicates the proportion of correctly predicted positive cases out of all cases predicted as positive.

- The recall, also known as sensitivity or true positive rate, is 40.30%. It signifies the proportion of actual positive cases that were correctly identified by the model.
- The F1-score is 0.50, which is the harmonic mean of precision and recall. It provides a balance between precision and recall, with higher values indicating better model performance.
- The ROC-AUC score is 0.9294, representing the area under the receiver operating characteristic curve. It measures the model's ability to distinguish between positive and negative classes, with higher values indicating better discrimination performance.

Overall, while the model exhibits high accuracy and ROC-AUC score, there is room for improvement in precision and recall, especially in correctly identifying positive cases.

9. Build a LDA Model on Train Dataset. Also showcase your model building approach

Model Output:

```
Confusion Matrix:
 [[576  37]
 [ 41  26]]
Classification Report:
              precision    recall  f1-score   support

           0       0.93      0.94      0.94       613
           1       0.41      0.39      0.40        67

    accuracy                           0.89       680
   macro avg       0.67      0.66      0.67       680
weighted avg       0.88      0.89      0.88       680
```

The model achieved an accuracy of 88.53% on the test dataset, with a confusion matrix indicating 576 true negatives, 26 true positives, 37 false negatives, and 41 false positives. While the model demonstrates high accuracy, precision and recall for class 1 (default cases) are relatively low, suggesting potential challenges in correctly identifying instances of default.

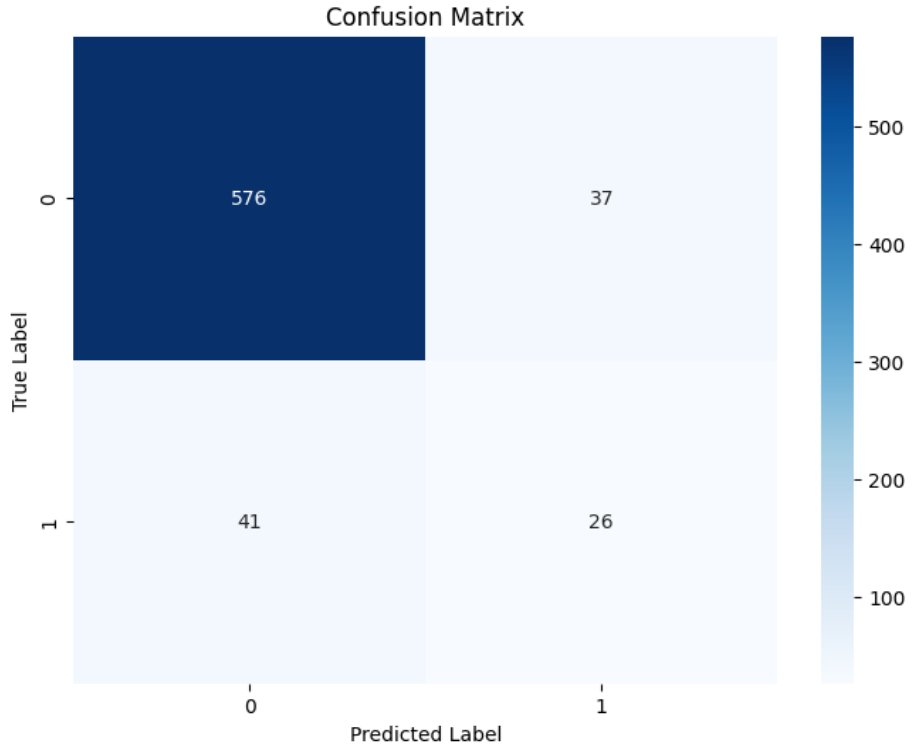


Figure 20: Confusion matrix for LDA Model

10. Validate the LDA Model on test Dataset and state the performance metrics. Also state interpretation from the model

Accuracy Output is

```
Performance Metrics for LDA:
Accuracy: 0.8955296404275996
Precision: 0.5172413793103449
Recall: 0.3409090909090909
F1-score: 0.410958904109589
ROC-AUC: 0.8790174596893857
```

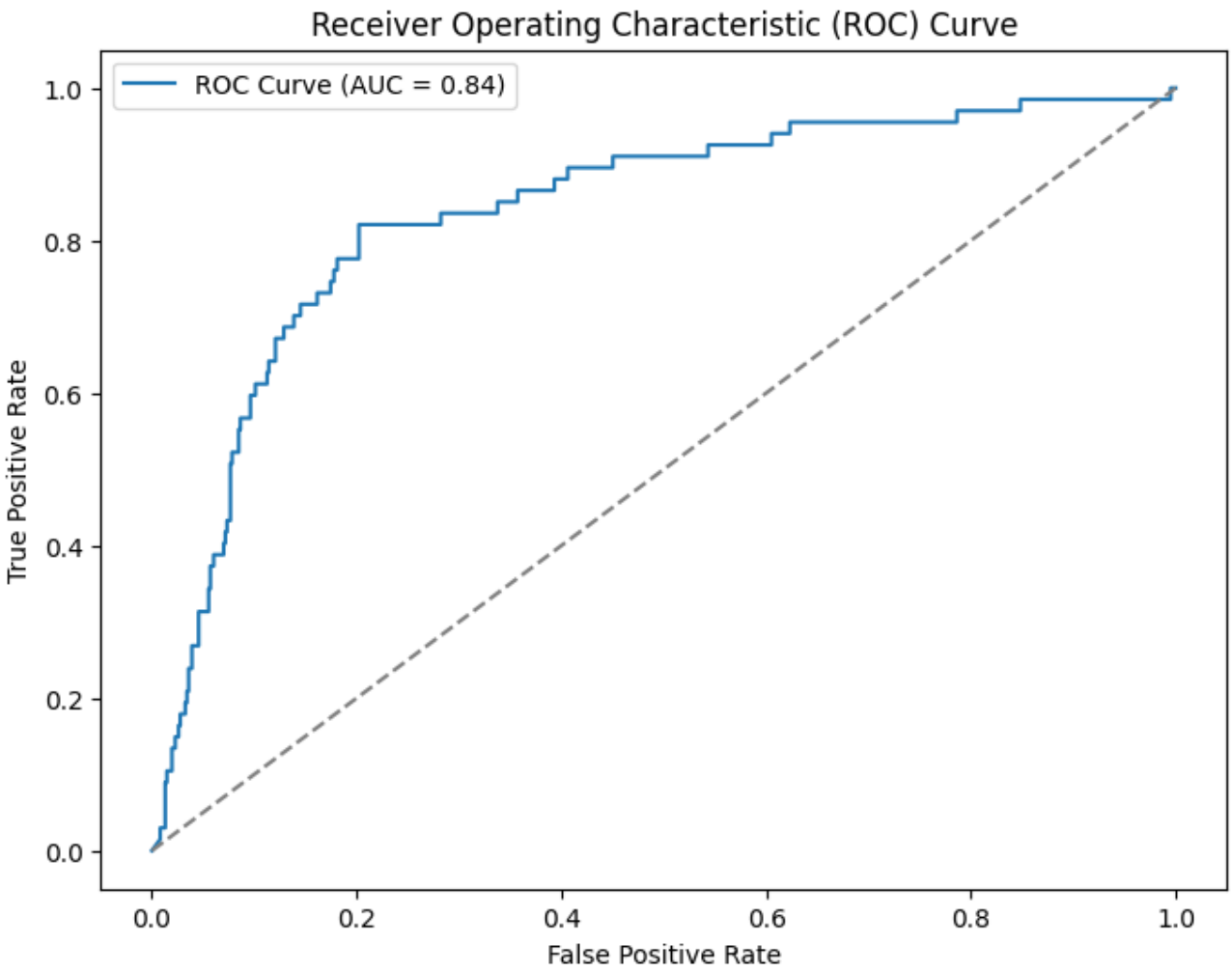


Figure 21: ROC curve for LDA Model

In the LDA model evaluation, the accuracy achieved was 89.55%, indicating the overall proportion of correctly classified instances. Precision, measuring the proportion of true positive predictions among all positive predictions, was 51.72%. Recall, which gauges the proportion of actual positives that were correctly identified by the model, stood at 34.09%. The F1-score, which balances precision and recall, was calculated at 41.10%. Finally, the ROC-AUC score, representing the area under the receiver operating characteristic curve, was 87.90%, indicating the model's ability to discriminate between default and non-default instances.

11. Compare the performances of Logistic Regression, Random Forest, and LDA models (include ROC curve)

- Comparing model outputs

```
Performance Metrics for Logistic Regres
Accuracy: 0.8661764705882353
Precision: 0.1
Recall: 0.04477611940298507
F1-score: 0.061855670103092786
ROC-AUC: 0.6079472133622263

Performance Metrics for Random Forest:
Accuracy: 0.9205882352941176
Precision: 0.6585365853658537
Recall: 0.40298507462686567
F1-score: 0.5
ROC-AUC: 0.9294270896739791
```

```
Performance Metrics for LDA:
Accuracy: 0.8852941176470588
Precision: 0.4126984126984127
Recall: 0.3880597014925373
F1-score: 0.4
ROC-AUC: 0.8377322198144677

Best Model: Random Forest
```

- Combined ROC curves for 3 models

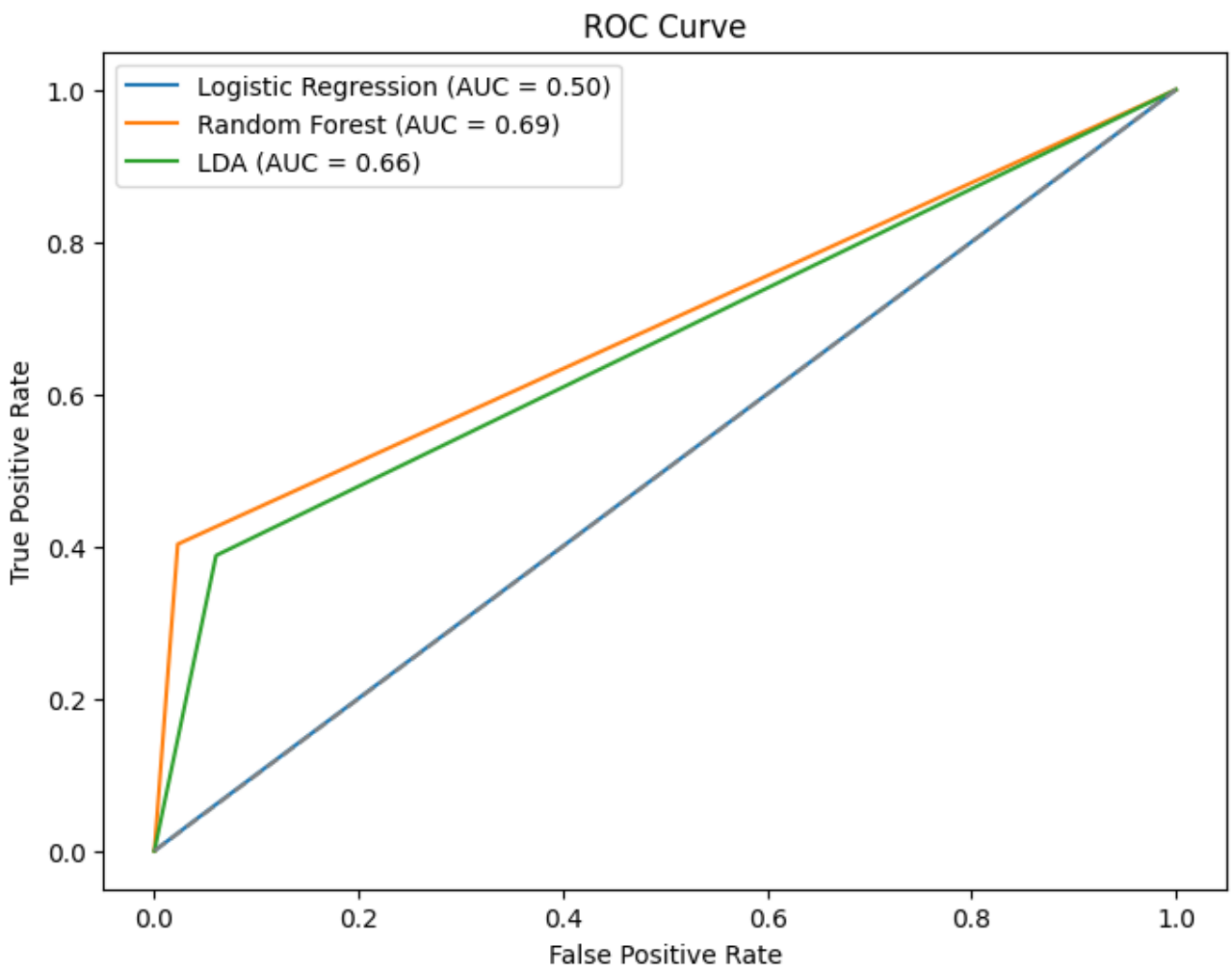


Figure 22: ROC curve for Logistic Regression, Random Forest and LDA Models

As per the analysis it has been concluded that Random-forest model is the best model with 92% accuracy.

12. Conclusions and Recommendations

Based on the performance metrics of the three models (Logistic Regression, Random Forest, and LDA), it is evident that the Random Forest model outperforms the others in terms of accuracy, precision, recall, F1-score, and ROC-AUC. Therefore, the Random Forest model is recommended for predicting defaults in this scenario.

Recommendations:

Utilize Random Forest Model: Given its superior performance across various metrics, the Random Forest model should be used for predicting defaults in the future. It provides better accuracy in identifying default cases and non-default cases, which is crucial for making informed business decisions.

Regular Monitoring: Despite the effectiveness of the Random Forest model, it's essential to continuously monitor its performance and recalibrate it as necessary. This ensures that the model remains accurate and reliable over time, especially as business conditions and data patterns may change.

Improve Data Quality: Enhancing the quality of input data can further improve the performance of predictive models. This includes ensuring data accuracy, completeness, and relevance. Additionally, exploring additional features or variables that may provide valuable insights into default prediction could be beneficial.

Risk Mitigation Strategies: Use the predictions from the Random Forest model to develop proactive risk mitigation strategies. Identify high-risk accounts or customers early on and implement measures to minimize the likelihood of defaults, such as offering alternative payment arrangements or reducing credit limits.

Strengthen Financial Management: Lastly, focus on strengthening financial management practices within the company. This includes optimizing cash flow, managing debt levels effectively, and diversifying revenue streams. A robust financial framework will not only reduce the risk of defaults but also enhance the overall financial health and resilience of the business.

Summary:

The Random Forest model emerges as the best-performing model for predicting defaults, surpassing Logistic Regression and LDA in accuracy, precision, recall, F1-score, and ROC-AUC. Leveraging this model can aid businesses in identifying potential default cases more accurately, enabling them to take proactive measures to mitigate risks and maintain financial stability. However, continuous monitoring, data quality improvement, and strategic risk management are essential for maximizing the effectiveness of the model and ensuring long-term business success.